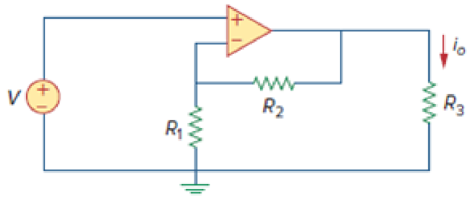
Chapter 5, Problem 26P

Problem

Using Fig. 5.64, design a problem to help other students better understand noninverting op amps.

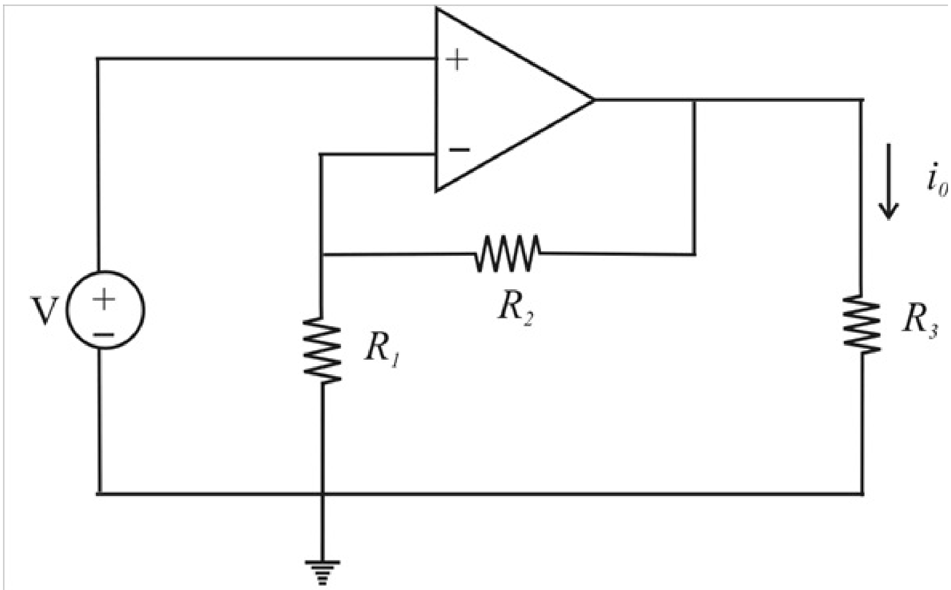
Figure. 5.64:



Step-by-step solution

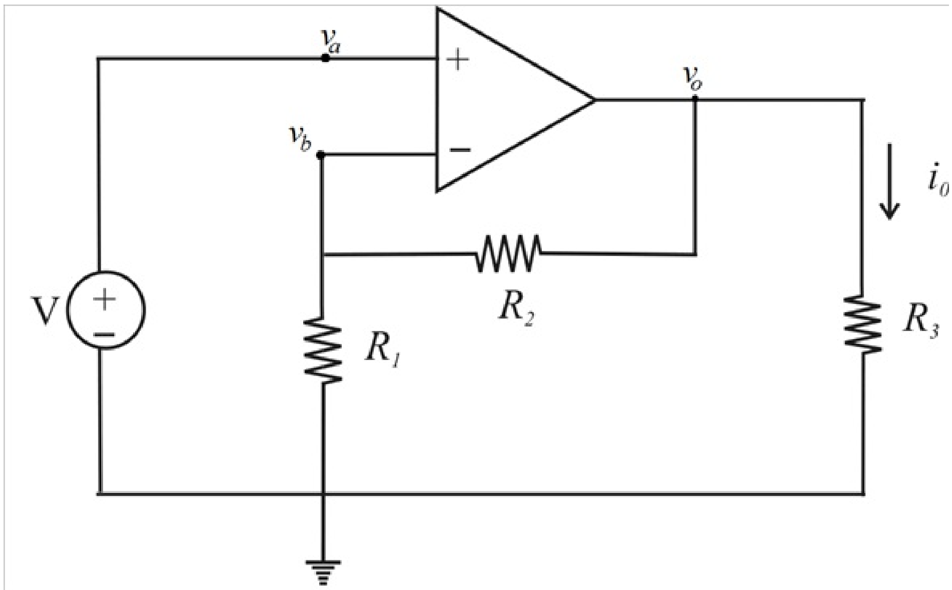
Step 1 of 5

Considering the non inverting OP amps



Step 2 of 5

For the analysis purpose the above circuit is modified as follows



Step 3 of 5

From the circuit $v_a = \mathrm{V}$

OP amp is ideal so

$$v_a = v_b$$

$$\therefore v_b = \mathrm{V}$$

Step 4 of 5

By applying KCL at node v_b

$$\frac{v_b}{R_1} + \frac{v_b - v_0}{R_2} = 0$$

$$v_b\left(\frac{1}{R_1} + \frac{1}{R_2}\right) = \frac{v_0}{R_2}$$

$$\mathrm{V}\left(\frac{R_1 + R_2}{R_1 R_2}\right) = \frac{v_0}{R_2}$$

$$v_0 = \mathrm{V}\left(\frac{R_1 + R_2}{R_1}\right)$$

Step 5 of 5

But $i_0 = \dfrac{v_0}{R_3}$

$$i_0 = \frac{\mathrm{V}\left(\frac{R_1 + R_2}{R_1}\right)}{R_3}$$

$$\therefore \boxed{i_0 = \mathrm{V}\left(\frac{R_1 + R_2}{R_1 R_3}\right) \mathrm{A}}$$